AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 25 · STUDENT LESSON

Generate, Retrieve, or Calculate?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Retrieve the facts. Calculate the result. Review the wording.

Trace mixed answers, inspect source versions and catch unsupported details without guessing methods from fluent text.

Keep one habit: Keep source facts → derived results → proposed wording distinguishable.

MAIN LESSON · SECTION 1

Which operation produced each part of the answer?

> Separate source facts, derived results and proposed wording.

Start with a mixed answer

A friendly invitation can mix a stored room number, a calculated number of places and newly proposed wording. Check each part using the evidence appropriate to the operation that produced it.

Recall the minimum necessary input

You need a chart of three fictional topic totals. Why should a named pupil spreadsheet not be the default **input**?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Your learning goals

- I can distinguish retrieving a record, calculating from **input**s and generating wording.
- I can **trace** a task that combines these operations and check each step.
- I can identify unsupported details and explain why **output** appearance does not reveal its method.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Missing and Unbalanced Examples. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Similarity. Give an example and a mistake that the idea helps you avoid.

Different systems create, look up, or compute answers in different ways.
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LEARNING TARGET Three tasks are: invent a school motto, find the exact fire-drill meeting point in a supplied handbook, and divide 84 stickers among 7 teams. Choose **generation**, **retrieval** or **calculation** for each.

Which **output**s need checking?

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can separate **retrieval**, **calculation** and **generate**d wording.

MAIN LESSON · SECTION 2

Meet the current makers-club notice

> Use the supplied fictional records, without an account or live search.

Define retrieval, calculation and generation

Three tasks can produce similar-looking text while doing different work. **Retrieval** finds existing information. **Calculation** applies mathematical operations to **input** values.

Generation produces content using a model and the context it receives. A single application can combine all three, so the label AI does not tell you which step happened.

Read N2 and its explicit replacement of N1

Our fictional makers club has a current notice N2: Wednesday, Room 4, six tables with four places at each table, and three places reserved for helpers. An older notice N1 says Room 2. N2 explicitly replaces N1 for this event.

These records are invented for this lesson; no search, account or real booking is needed.

Match each question to its operation

To answer “Which room does the current notice list?”, find N2 and return Room 4 with the record identifier. That is **retrieval**. To answer “How many places remain after the helper reservation?”, use 6 × 4 − 3 = 21.

That is **calculation**. To suggest a friendly invitation, **generate** wording and check that its factual details agree with N2.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A story generator creates new wording, a library search **retrieve**s a stored title, and a calculator computes a result from stated operations.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

MAIN LESSON · SECTION 3

Trace facts, arithmetic and wording separately

> Check the source version and whether the expression answers the actual question.

Follow the three-operation trace

One club task with three kinds of work

OPERATION	EXAMPLE OUTPUT	REQUIRED CHECK
Retrieve N2	Room 4 on Wednesday	Right event and version
Calculate	6 × 4 − 3 = 21	Inputs and operations
Generate wording	Try a making activity	Facts and constraints

Retrieval supplies recorded information; calculation derives a result; generation drafts wording. Each operation needs a check suited to what it does.

Inspect which notice governs this task

Retrieving N1 accurately would still give an outdated room for this task. Successful **retrieval** means finding a record, not proving that every record is correct or current. Check the event, source and version.

Here the explicit replacement statement tells us which notice governs the classroom example.

Distinguish total from remaining places

- A calculator given 6 × 4 correctly returns 24, but 24 is the total capacity, not the number remaining after three reserved places. Correct arithmetic on an incomplete expression answers the wrong question.

  Explain the **input** values, units and operations before accepting a number.

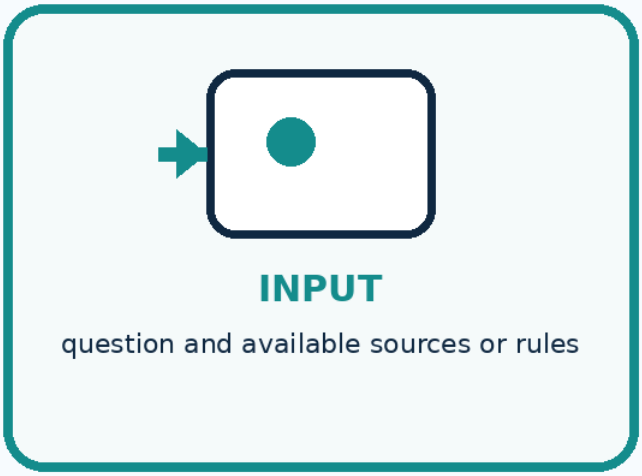
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

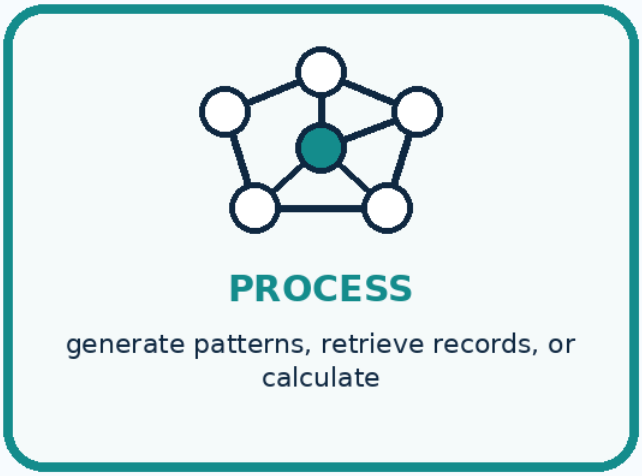
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VISUAL **MECHANISM**

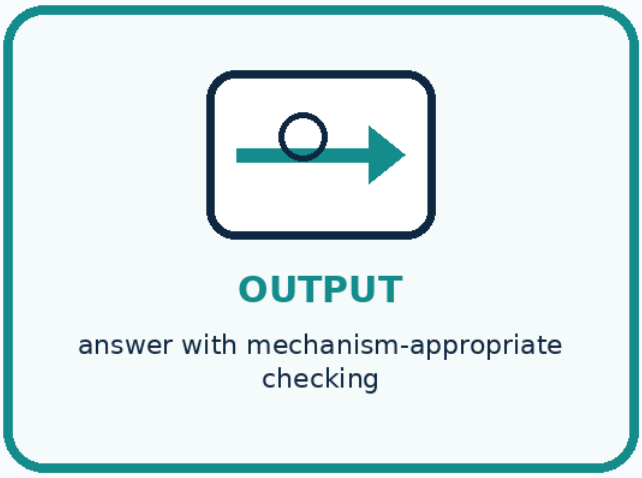
Input → Process → Output

Follow the information instead of imagining magic inside the machine.





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INPUT question and available sources or rules

PROCESS generate patterns, **retrieve** records, or **calculate**

OUTPUT answer with **mechanism**-appropriate checking

CORE IDEA Different systems create, look up, or compute answers in different ways.

UNDER THE HOOD **Generation** synthesizes **output**s from learned distributions, **retrieval** selects stored records, and **calculation** applies formal operations.

Products often combine all three; verification should match the component that produced each claim.

Draw a fresh example with all three stages:

A closer explanation

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare operations and their checks

> Similar-looking text can have different evidence paths.

Pictures paired with their evidence



The case to investigate

CASE 1 — Invent a story Clue: new wording from patterns. Meaning: **generate**.



The case to investigate

CASE 2 — Find a book title Clue: return catalog record. Meaning: **retrieve**.



The case to investigate

CASE 3 — Add 27 + 18 Clue: apply arithmetic. Meaning: **calculate**.



The case to investigate

CASE 4 — Answer with quote and total Clue: multiple components. Meaning: mixed system.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

BOUNDARY Fluent wording does not reveal whether a fact was **retrieve**d, **generate**d, or **calculate**d.

▼ Earlier wording, alternatives and picture transcripts

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Invent a story Clue: new wording from patterns. Meaning: **generate**.

CASE 2 — Find a book title Clue: return catalog record. Meaning: **retrieve**.

CASE 3 — Add 27 + 18 Clue: apply arithmetic. Meaning: **calculate**.

CASE 4 — Answer with quote and total Clue: multiple components. Meaning: mixed system.

Change one thing in the pictured case

Change: Reserve four places instead of two, keeping the room and day unchanged.

Prediction: Eight learner places remain.

Reason: 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

Trace the separate craft invitation

> Use approved record C and keep unsupported snack claims out of the draft.

The approved record and reservation

A fictional craft invitation combines three different operations.

Approved record C: Tuesday, Hall 4, three tables with four total places each.

Two places are reserved for helpers; no snack information is supplied.

Return to the makers-club notice N2

AI ACADEMY / AI-1/25 · 03 · INSPECT THE LESSON MODEL

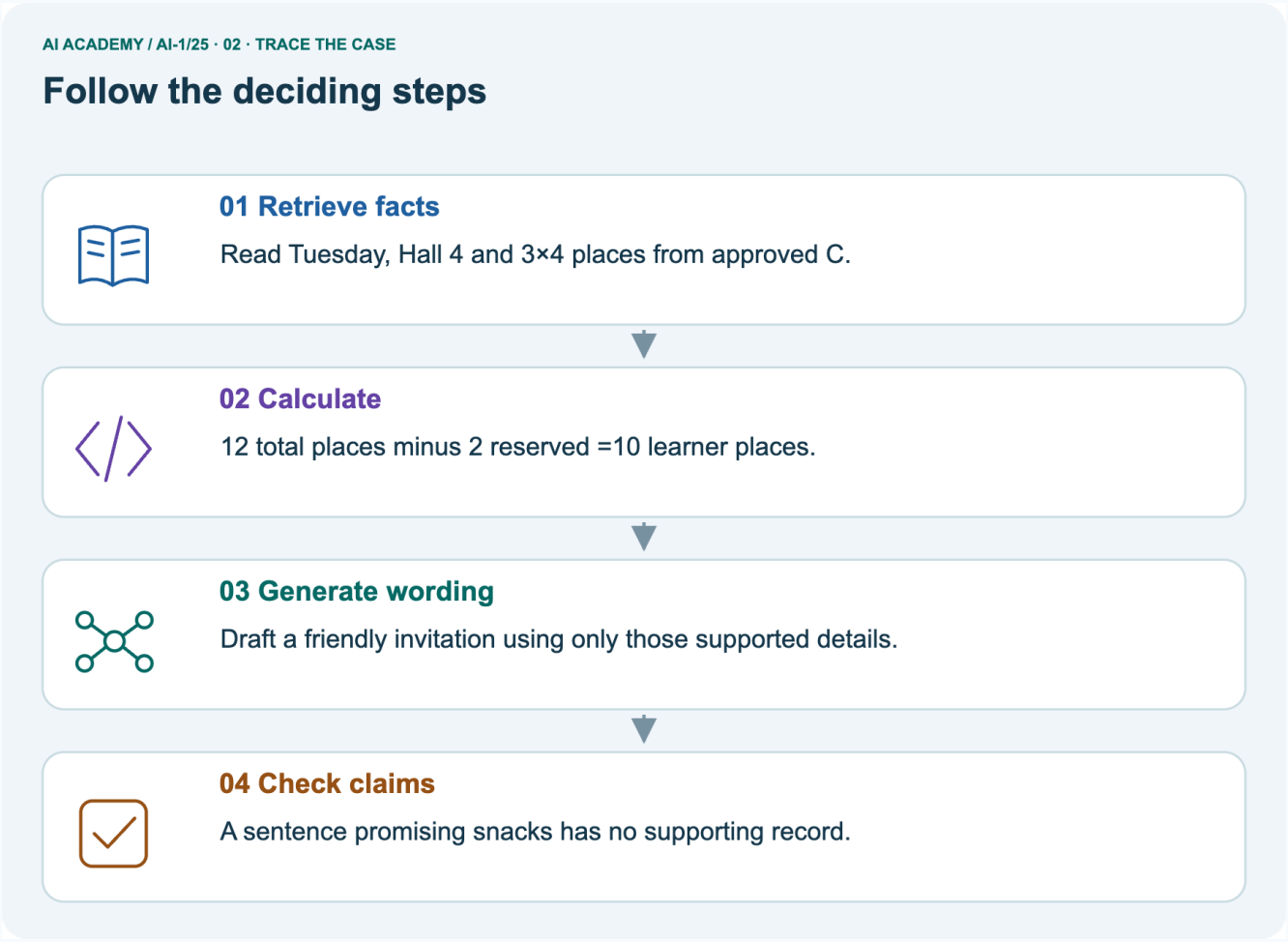
One club task with three kinds of work

Operation	Example output	Required check
Retrieve N2	Room 4 on Wednesday	Right event and version
Calculate	6 × 4 − 3 = 21	Inputs and operations
Generate wording	Try a making activity	Facts and constraints

Retrieval supplies recorded information; calculation derives a result; generation drafts wording. Each operation needs a check suited to what it does.

Follow the labelled picture with the matching explanation. Open the vector diagram

Follow retrieval, calculation and wording



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Retrieve facts	Read Tuesday, Hall 4 and 3×4 places from approved C.
2. Calculate	12 total places minus 2 reserved =10 learner places.
3. Generate wording	Draft a friendly invitation using only those supported details.
4. Check claims	A sentence promising snacks has no supporting record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Explain why each claim needs its own check

- **Retrieval** copies relevant source information; **calculation** follows arithmetic; **generation** proposes wording. These jobs can cooperate, but one does not validate the others.

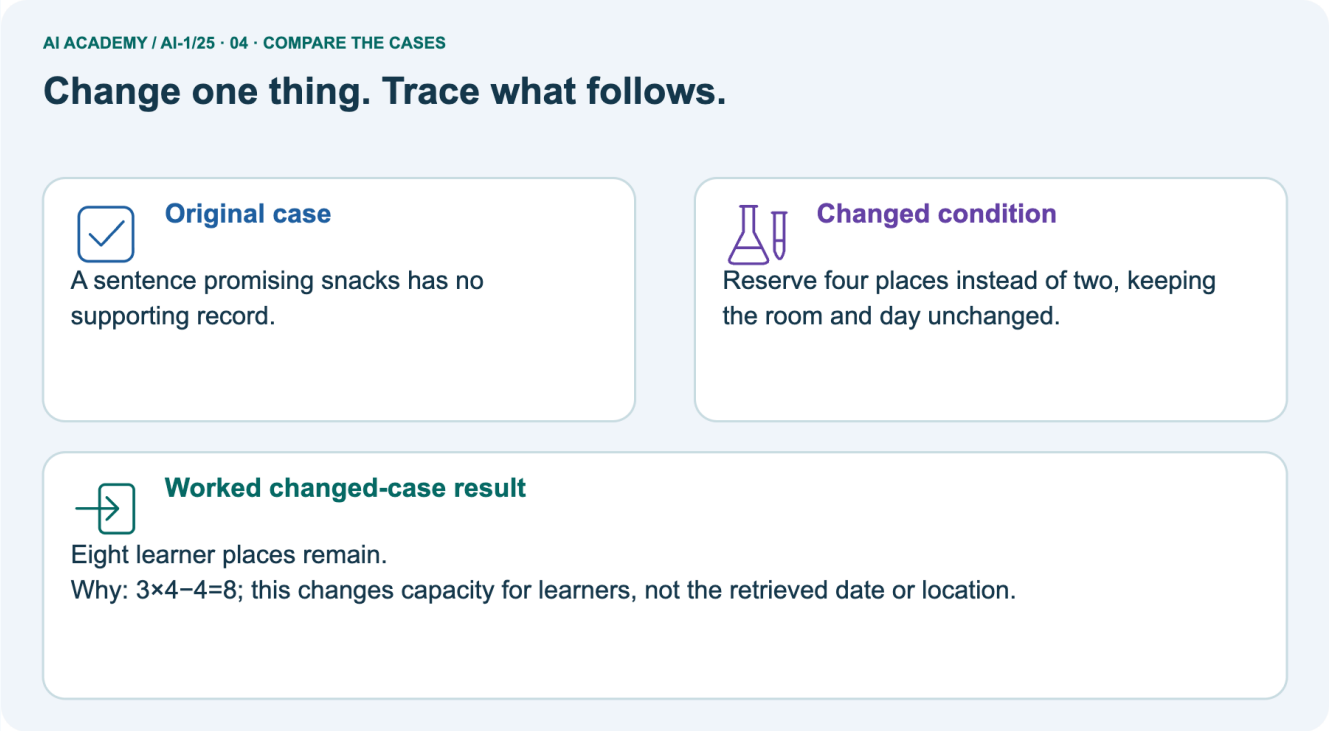
  A friendly **generate**d sentence can still contain an invented detail. Keep the approved source beside the draft and check each factual claim separately.

- Read the labelled picture: Label each arrow with its operation. Can the system add snacks because invitations often mention them?

Predict a reservation change

- Before opening the changed-reservation picture, reserve four of the twelve total places for helpers instead of two. Predict the remaining learner places and name the retrieved facts that do not change.

▼ Check the changed calculation



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit what retrieval establishes

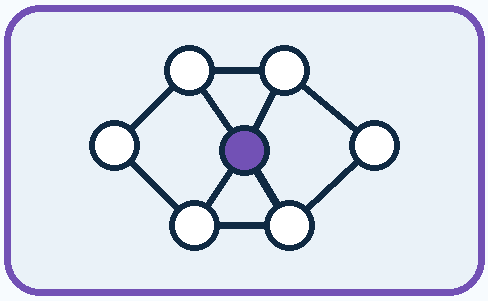
The fictional source must itself be current and authoritative; **retrieval** alone does not ensure a record is correct.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A story generator creates new wording, a library search **retrieve**s a stored title, and a calculator computes a result from stated operations.

STEP 1 — OBSERVE question and available sources or rules

STEP 2 — **TRACE generate** patterns, **retrieve** records, or **calculate**

STEP 3 — CONCLUDE answer with **mechanism**-appropriate checking

MODEL REASONING Check **generate**d claims against independent **evidence**, check **retrieve**d facts against the cited record and freshness, and recompute **calculation**s or inspect the operation. Mixed **output**s need more than one check.

DO NOT OVERCLAIM Fluent wording does not reveal whether a fact was **retrieve**d, **generate**d, or **calculate**d.

What new **evidence** could change this answer?

A closer explanation

Draft statement	Evidence in this example	Decision
Meet Wednesday in Room 4	Current notice N2	Supported by the record
There are 21 non-helper places	6 × 4 − 3 from N2	Supported by calculation

Draft statement	Evidence in this example	Decision
Free snacks will be provided	Not stated in N2	Remove or verify separately

A **generate**d invitation might add free snacks because that detail fits a familiar invitation pattern. Plausible wording is not **evidence** that snacks are provided.

Remove that claim or obtain an appropriate confirming source before including it. **Generate**d content can be useful and can contain correct facts; its fluent style alone does not verify those facts.

Generation also does not guarantee originality. A response may repeat wording already present in its context or elsewhere.

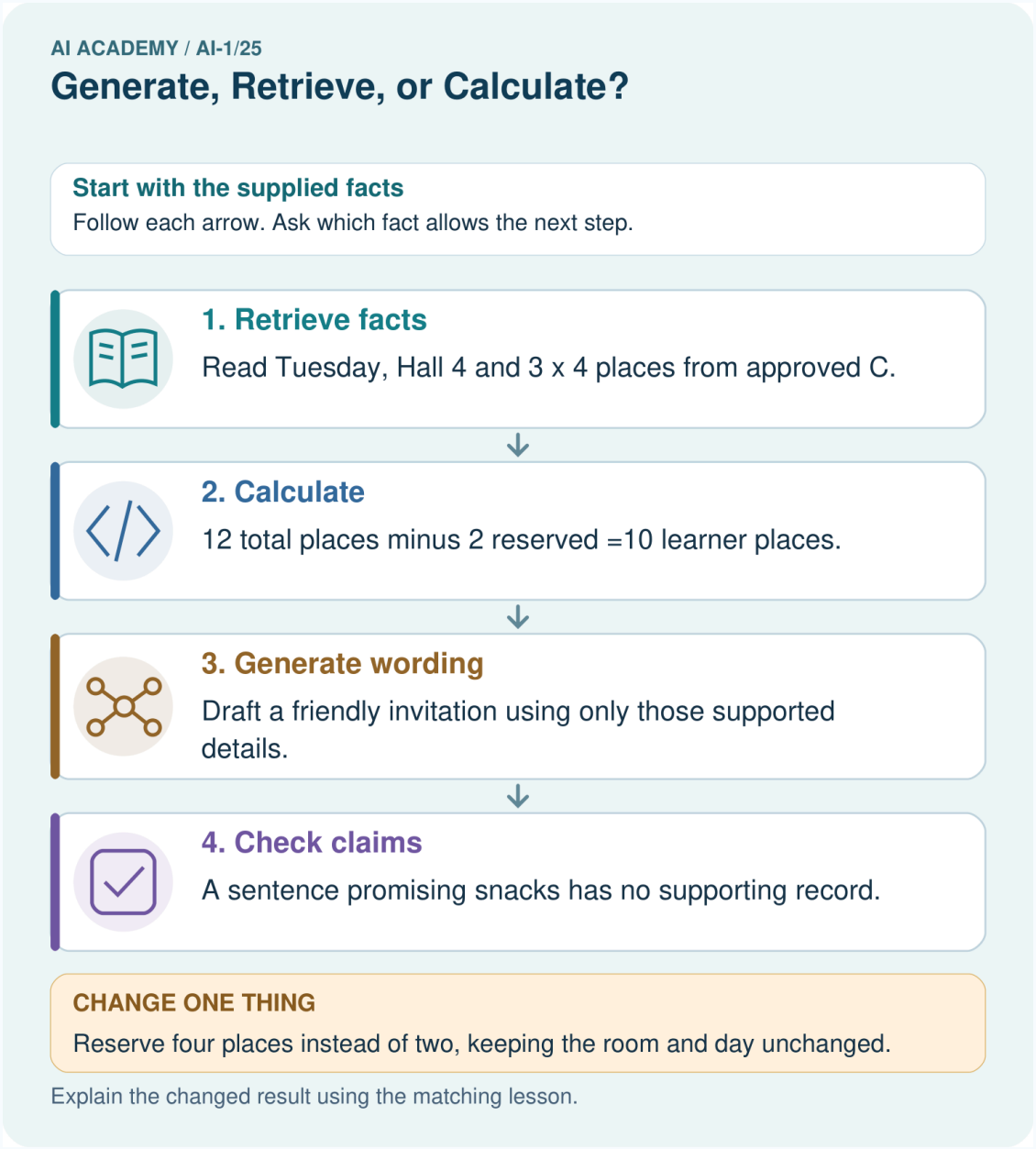
Retrieval and **calculation** can feed a generator, and a generator can propose a **calculation** for another tool. Keep the **evidence** path visible instead of assuming that every sentence came from the same operation.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Generate, Retrieve, or Calculate?

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing: Reserve four places instead of two, keeping the room and day unchanged. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Choose a method because it fits the question

> Make the evidence path inspectable and retain unknown facts as unknown.

Match the operation to the job

For an existing meeting time, **retrieve** the relevant notice. For a total, **calculate** using the right **input**s and units. For a draft title or invitation, **generation** can offer wording, followed by review.

You can also do these classroom tasks on paper. Choose the operation because it fits the question, not because a tool is described as intelligent.

Write an answer that can be updated

A complete answer keeps source facts, derived results and suggested wording distinguishable. It marks missing facts instead of inventing them, preserves the source identifier and explains the arithmetic.

This makes a later change easier to handle: if the number of reserved places changes, recompute the result and update the sentence.

Repair both tempting claims

- CLAIM A Any answer on a screen came from a database.
- CLAIM B **Calculate**d **output**s never need checking.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

▼ **Earlier wording, alternatives and picture transcripts**

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Invent a story	new wording from patterns	generate
Find a book title	return catalog record	retrieve
Add 27 + 18	apply arithmetic	calculate
Answer with quote and total	multiple components	mixed system

Repair two tempting claims

Repair:

Repair:

A closer explanation

> **If N2 now reserves five helper places, calculate the remaining places. State which existing source facts stay the same and why the old answer of 21 must be revised**

MAIN LESSON · SECTION 8

Practise: matching answers do not establish matching methods

> Inspect the record and recompute rather than accepting “I checked” as proof.

Compare the two displayed answers

Two applications display “21 places remain.” One **trace** shows **retrieval** of N2 followed by a calculator call for 6 × 4 − 3 and a sentence built from the result. The other gives no record or tool **trace**.

Their identical visible answers do not establish identical methods. The second answer may be right, but its appearance does not prove **retrieval** or **calculation** took place.

Check independently and state what remains unknown

Check the record yourself and independently recompute 6 × 4 − 3. A displayed **trace** is useful **evidence**, but it should refer to an inspectable record and the actual tool result.

A sentence saying “I checked” is not the same as a check you can inspect. If **evidence** is unavailable, say what remains unverified.

Repair an incomplete expression in W2

Use workbook W2 for the connected example “**Generation retrieval** and **calculation**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your **trace** to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge the current-record claim

- Use the worked case for **Generation retrieval** and **calculation**. Proposed explanation: “Retrieving a record proves it is current”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the mixed craft invitation

Label each arrow with its operation. Can the system add snacks because invitations often mention them?

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Different systems create, look up, or compute answers in different ways.

Use a fresh example to explain **generate**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: **Calculate**d **output**s never need checking.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

> **A draft says “Join us Thursday in Room 4 for free snacks; 24 places remain.” Use N2 to identify which claims need correction or removal. Write a corrected factual sentence and state the retrieval and calculation steps that support it**

Practise with less help: Separate a correct sum from a correct answer

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Investigate nine operation cards

> Use supplied descriptions rather than guessing a method from an answer’s appearance.

Read the nine station tasks

Nine paper station cards describe operations; classify each described step, then name its check. Cards 1–3: copy the day from current fictional notice J (Saturday); return the room from J (Room 6); return a title from an approved catalogue entry K (Paper Bridges). Cards 4–

6: compute 4×5 total places; subtract 3 helpers from 20 places; divide 18 cards equally among 3 tables. Cards 7–9: propose a friendly workshop title; draft a two-sentence invitation using J; suggest a fictional character for a story. J contains no snack or transport promise.

Classify, trace and check each operation

1. For each card, predict the main operation and record its input, result or proposed output, and suitable check. On the calculation cards, show intermediate work and units. On wording cards, distinguish suggestions from factual claims and compare factual details with J. Then combine one source fact, one calculation and one proposed sentence into an inspectable answer. These are supplied paper traces, not evidence that an AI tool was run.
2. Keep a normal case and an edge case clearly separated.
3. Record what happened—even when it does not match your **prediction**.
4. Explain the result, one limitation, and the next useful test.

Record evidence and limitations

Case	Prediction	Observed result	What it means
Normal			
Edge			

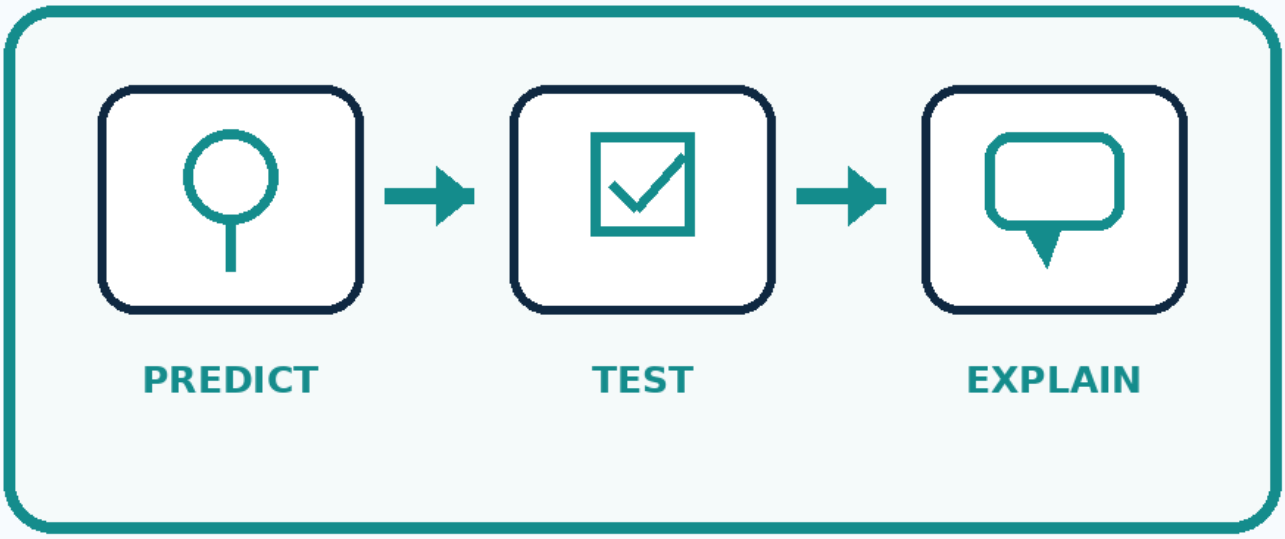
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only **trace**d the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Mechanism Sorting Station

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Classify nine **output**s by **mechanism** and verification need.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: trace the new library workshop

> Use record B, then calculate and repair the draft before feedback.

The independent record and draft

Current fictional record B lists a library workshop on Friday in Room 5: five tables with six places each, with four places reserved for helpers. B replaces old record A, which lists Room 3.

A draft says “Thursday in Room 3, with 30 places remaining and a free bus home.” **Retrieve** the current day and room, **calculate** the non-helper places, write a corrected factual sentence and identify the detail that cannot be supported by B.

Keep your first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Trace a new library workshop

My independent explanation, trace and limits

Week 25 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Choose a method for each task: invent a dragon story; find the library's posted hours; compute 7 plus 5. Explain why the methods differ.

My answer, with a **trace**, **calculation** or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Label every sorter component as **generate**, **retrieve**, **calculate**, fixed rule, learned **prediction**, or **human decision**.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

Three tasks are: invent a school motto, find the exact fire-drill meeting point in a supplied handbook, and divide 84 stickers among 7 teams. Choose **generation**, **retrieval** or **calculation** for each. Which **output**s need checking?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Label the sorter’s actual components

> Use the broader component labels where retrieval, calculation and generation do not fit.

This week’s project decision

THIS WEEK'S CHECKPOINT Label every sorter component as **generate**, **retrieve**, **calculate**, fixed rule, learned **prediction**, or **human decision**.

Use the sorter’s approved catalogue for facts, explicit **calculation**s for scores and checked prose for explanations.

Inspect inputs, sources and uncertain outputs

- What real classroom need does this serve?
- What information is necessary—and what should stay private?

- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

Keep appearance separate from method

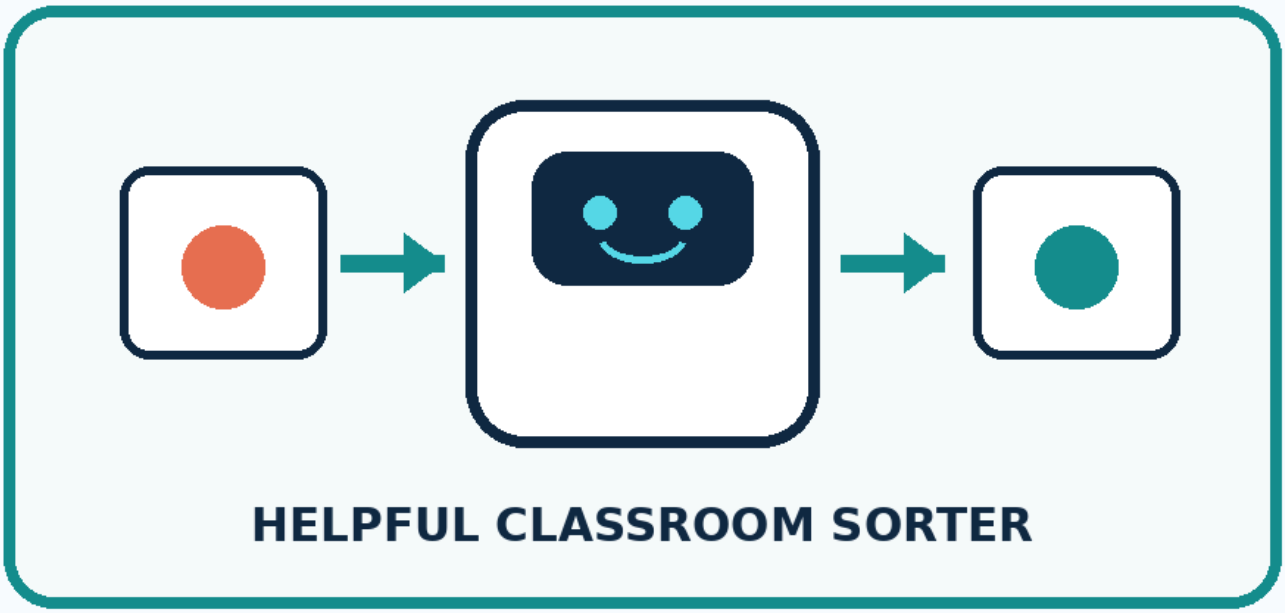
RESPONSIBLE DESIGN Fluent wording does not reveal whether a fact was **retrieve**d, **generate**d, or **calculate**d.

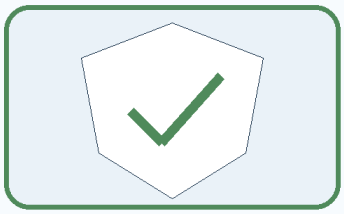
▼ Earlier wording, alternatives and picture transcripts

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain an answer's evidence path

> Close your notes and name an appropriate check for each operation.

Your exit explanation

- Close your notes. Explain this goal in your own words: Distinguish retrieving a record, calculating from **input**s and generating wording

> **My deciding step, evidence and one remaining question**

Choose a changed-case check

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
generate	create a new output from learned patterns	
retrieve	find and return stored information	
calculate	apply defined operations to values	
provenance	where an output or claim came from	

Terms for the connected explanation

Word	Meaning
Retrieval	Finding and returning existing information from a collection
Calculation	Applying specified mathematical operations to input values
Generation	Producing content using a model and its input context

Word	Meaning
Source record	An existing item whose content and origin can be inspected
Trace	A record of the inputs, operations and outputs used in a task

Repair a claim using a word

- CLAIM TO REPAIR Any answer on a screen came from a database.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

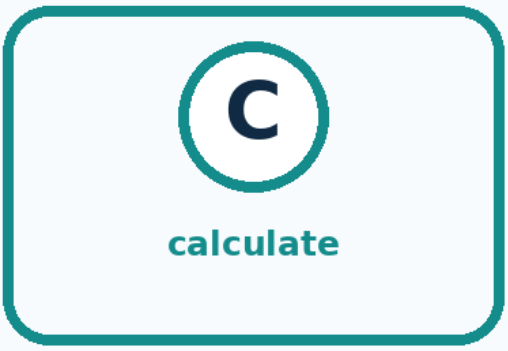
VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

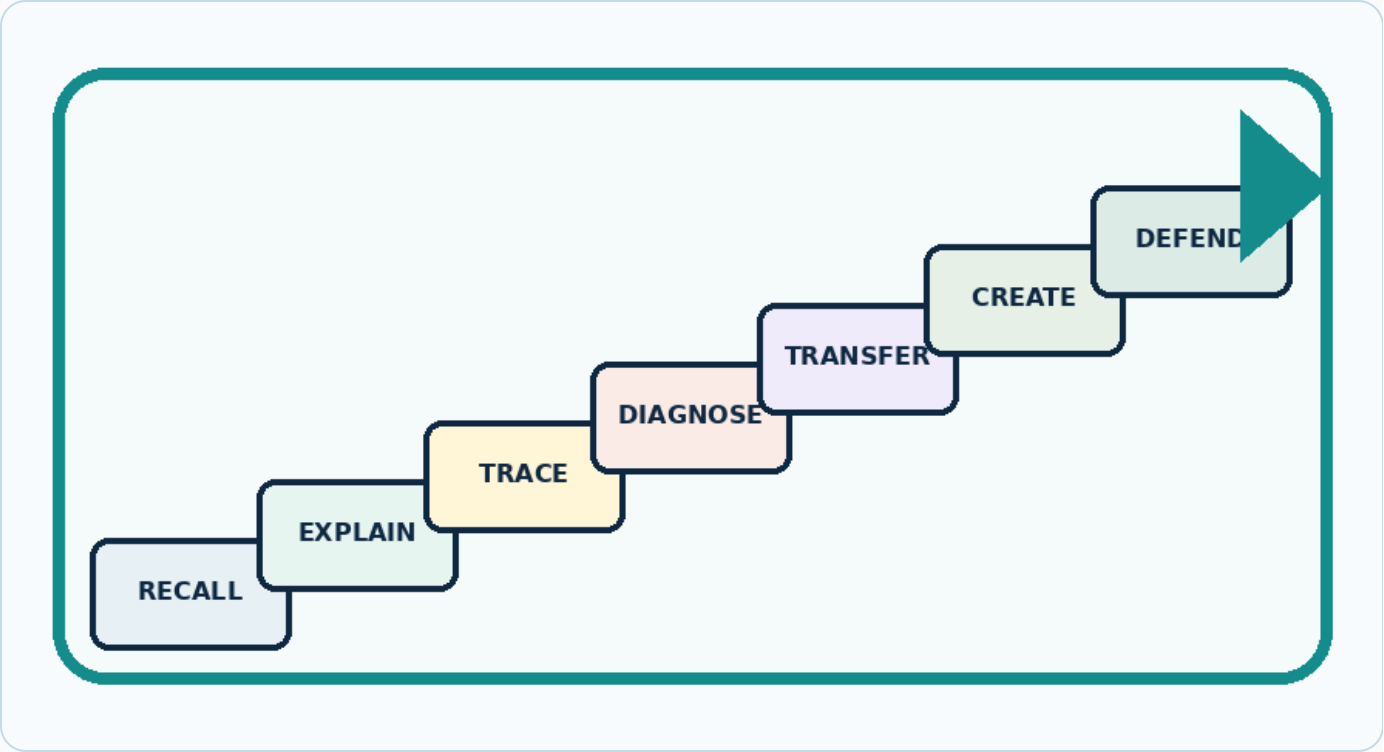
- RECALL Define **generate**.
- EXPLAIN Different systems create, look up, or compute answers in different ways.
- **TRACE** Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: Any answer on a screen came from a database.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Trace a new library workshop is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ **Trace input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Trace a new library workshop is the selected independent case.



EXPLANATION CHECK **Retrieve** the governing record, use a complete expression and check each factual claim in the proposed wording.

Independent **evidence**:

SELF-CHECK Check **generate**d claims against independent **evidence**, check **retrieve**d facts against the cited record and freshness, and recompute **calculation**s or inspect the operation. Mixed **output**s need more than one check.

NEXT BRIDGE Carry today's **evidence** habit into: Prompts Are Design Instructions.

▼ **Extra practice: Choose the operation that fits**

REFERENCE · SECTION 13

Extra practice: Choose the operation that fits

> Try the question before opening its worked explanation.

Read the supplied case and question

A learner needs a new poem, 18+7, and a passage from the class handbook.

Choose the operation for each.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Generate the poem, **calculate** 25, **retrieve** the actual passage. Inventing a fluent passage is not **retrieval**.

Evidence or stage	Value or result
New rain poem	Generate
18 + 7	Calculate → 25
Handbook passage	Retrieve the actual text

▼ Earlier wording, alternatives and picture transcripts

Week 25 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Choose the operation that fits

A learner needs a new poem, 18+7, and a passage from the class handbook.

Choose the operation for each.

Generate the poem, **calculate** 25, **retrieve** the actual passage. Inventing a fluent passage is not **retrieval**.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Distinguish retrieving a record, calculating from inputs and generating wording	Explain the deciding step, show a trace or test, and state one limit.
Trace a task that combines these operations and check each step	Explain the deciding step, show a trace or test, and state one limit.
Identify unsupported details and explain why output appearance does not reveal its method	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional craft invitation combines three different operations.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional craft invitation combines three different operations.
- Which labelled step matches this detail? A sentence promising snacks has no supporting record.
- 1. **Retrieve** facts
- 2. **Calculate**
- 3. **Generate** wording
- 4. Check claims
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

After your attempt, compare with the supported explanation: Check claims A sentence promising snacks has no supporting record.

Retrieval copies relevant source information; **calculation** follows arithmetic; **generation** proposes wording. These jobs can cooperate, but one does not validate the others.

A friendly **generate**d sentence can still contain an invented detail. Keep the approved source beside the draft and check each factual claim separately.

Changed-case question

- Changed-case practice: Reserve four places instead of two, keeping the room and day unchanged. Which conclusion is supported by the supplied facts?
- 1. Eight learner places remain.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial **trace**. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: Eight learner places remain. 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

Repair and save your work

Review Privacy by Choosing Less (ai-1/24). Goal: Justify which information a stated task needs and remove unnecessary fields.

Short practice: For the original task, does adding an anonymous-looking pupil code improve the repair count?. Return to this week and explain what you changed.

Review Missing and Unbalanced Examples (ai-1/22). Goal: Name the target collection, selection list and cases actually assessed. Short practice: Would testing only dim plain cards close every current gap?

Point to the target groups.. Return to this week and explain what you changed.

Review Similarity (ai-1/19). Goal: **Calculate** similarity using explicitly selected features and a stated scoring rule. Short practice: Point to the matches supporting C's new score and explain why there is no tie..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

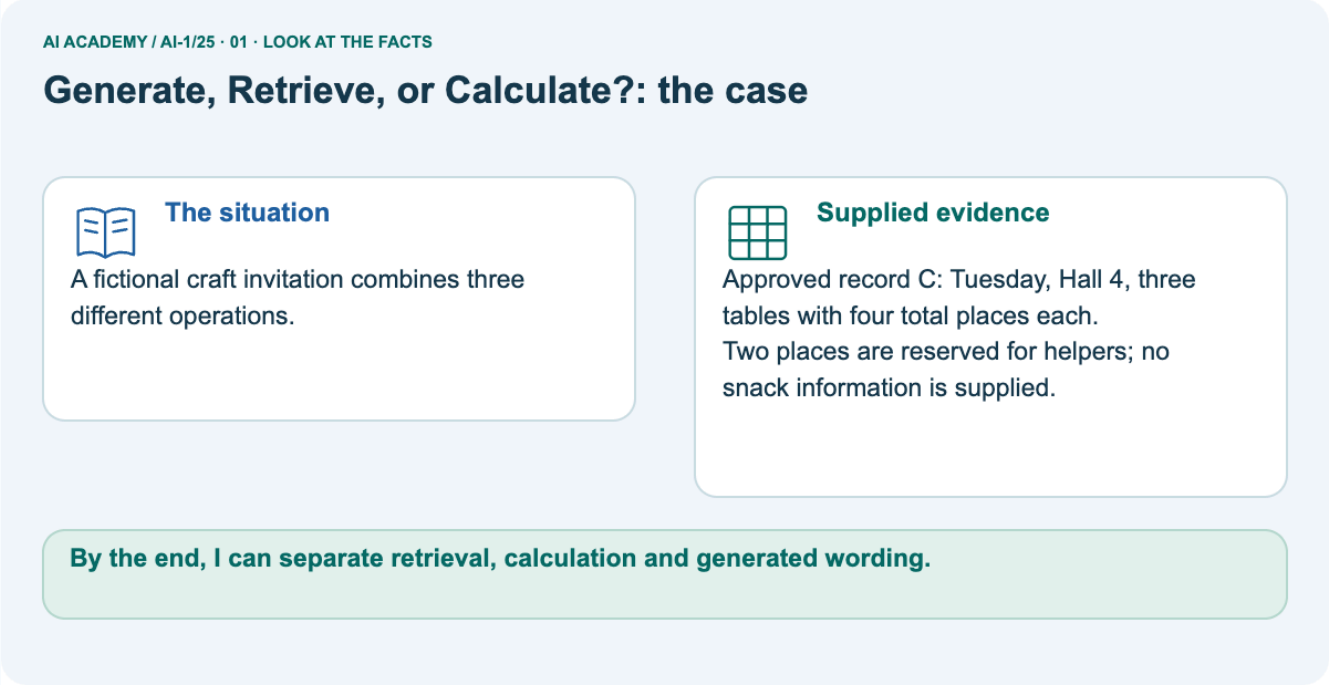
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Generate, Retrieve, or Calculate?: the case



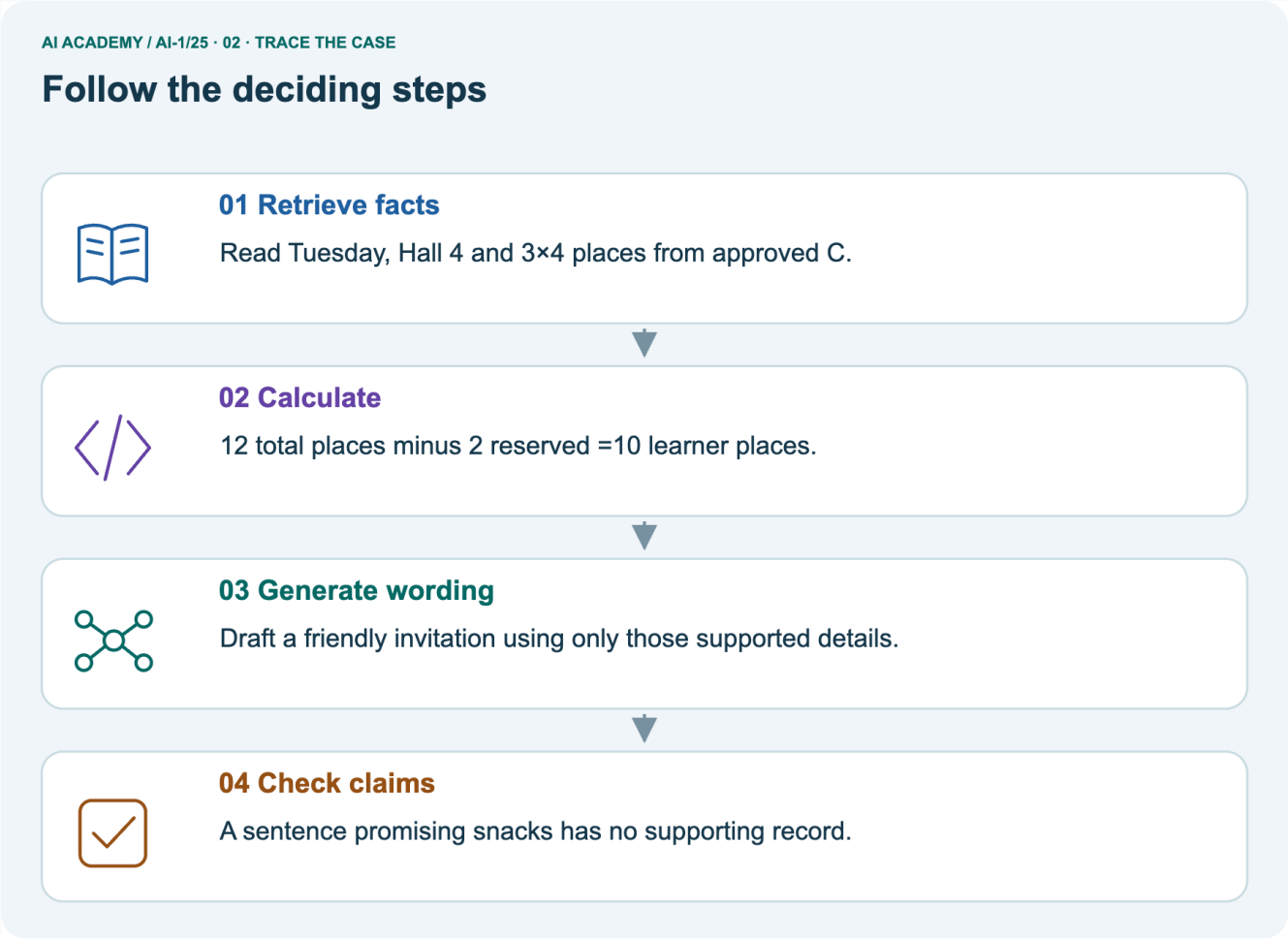
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional craft invitation combines three different operations. Supplied facts: Approved record C: Tuesday, Hall 4, three tables with four total places each.

Two places are reserved for helpers; no snack information is supplied.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

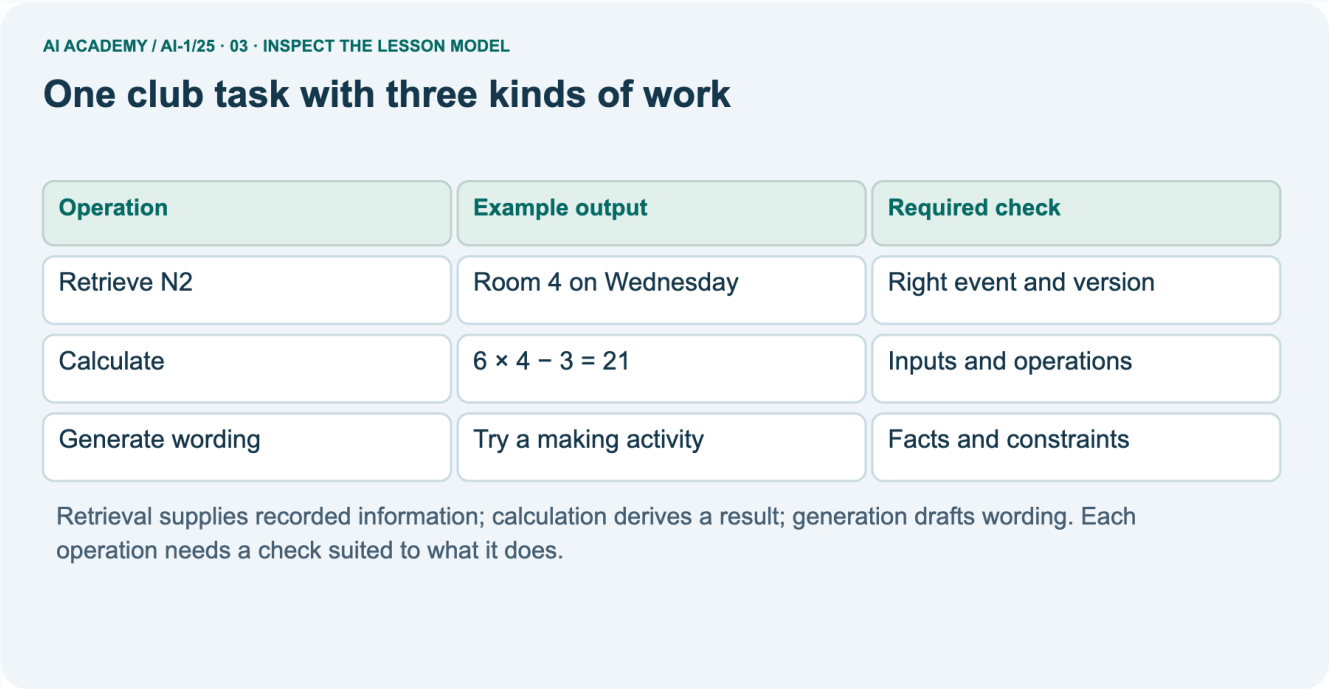


Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve facts: Read Tuesday, Hall 4 and 3×4 places from approved C. → **Calculate**: 12 total places minus 2 reserved =10 learner places. → **Generate** wording: Draft a friendly invitation using only those supported details. → Check claims: A sentence promising snacks has no supporting record.

Point to each numbered stage and explain which recorded **input** it uses.

03. One club task with three kinds of work



AI ACADEMY / AI-1/25 · 03 · INSPECT THE LESSON MODEL

One club task with three kinds of work

Operation	Example output	Required check
Retrieve N2	Room 4 on Wednesday	Right event and version
Calculate	6 × 4 − 3 = 21	Inputs and operations
Generate wording	Try a making activity	Facts and constraints

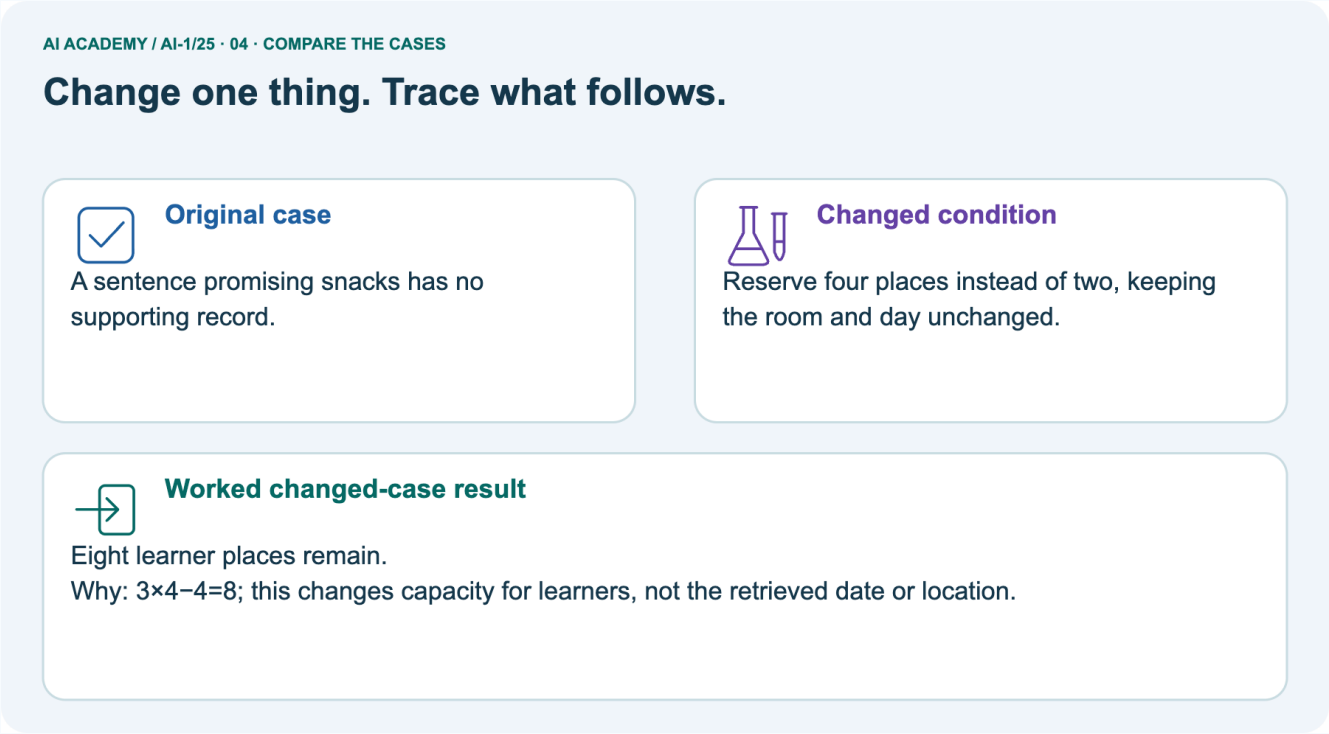
Retrieval supplies recorded information; calculation derives a result; generation drafts wording. Each operation needs a check suited to what it does.

Follow the labelled picture with the matching explanation. Open the vector diagram

One club task with three kinds of work. **Retrieve** N2 | Room 4 on Wednesday | Right event and version. **Calculate** | 6 × 4 − 3 = 21 | **Input**s and operations. **Generate** wording | Try a making activity | Facts and constraints

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



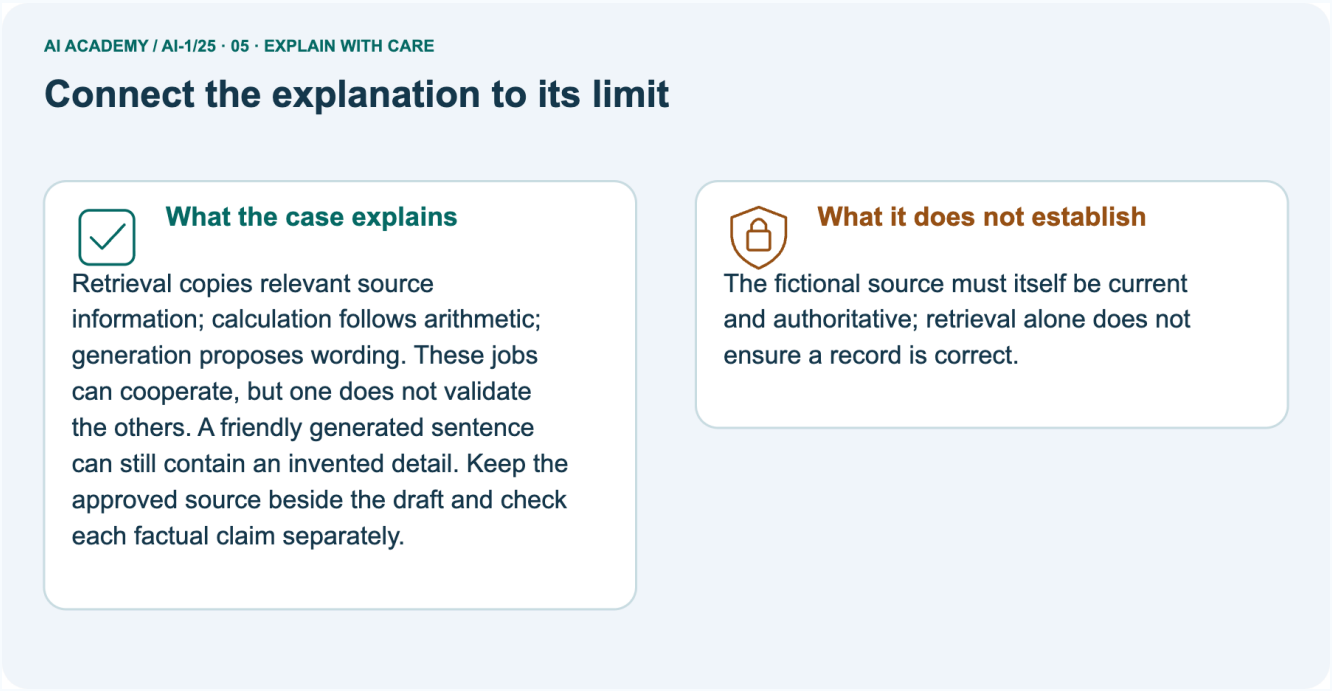
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A sentence promising snacks has no supporting record. Changed condition: Reserve four places instead of two, keeping the room and day unchanged. Expected result: Eight learner places remain.

Reason: 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

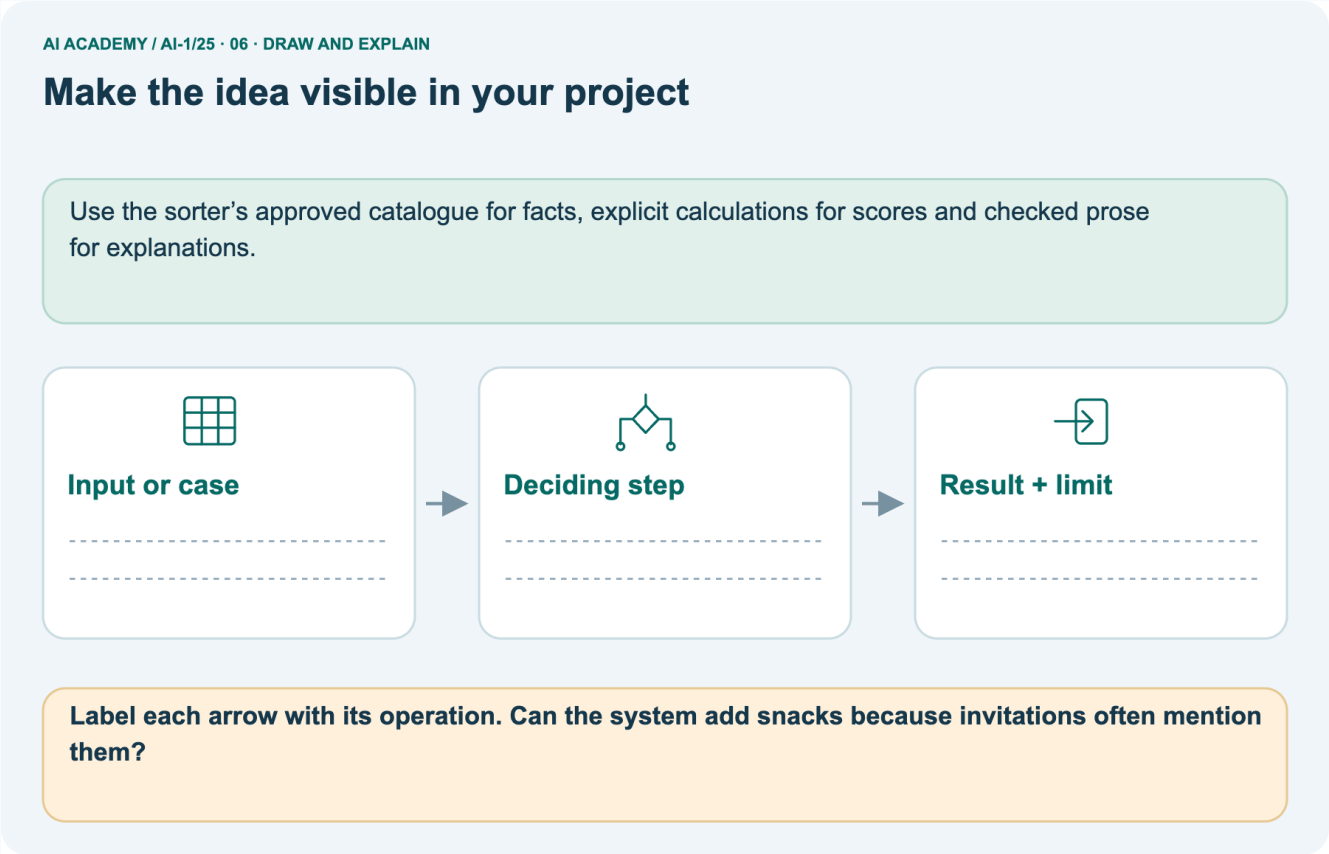
Retrieval copies relevant source information; **calculation** follows arithmetic; **generation** proposes wording. These jobs can cooperate, but one does not validate the others.

A friendly **generate**d sentence can still contain an invented detail. Keep the approved source beside the draft and check each factual claim separately.

Limit: The fictional source must itself be current and authoritative; **retrieval** alone does not ensure a record is correct.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use the sorter's approved catalogue for facts, explicit **calculation**s for scores and checked prose for explanations. Draw your **input**, deciding step, result and limit. Supported question: Label each arrow with its operation.

Can the system add snacks because invitations often mention them?

Draw your project **trace** and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Generate, Retrieve, or Calculate?: the case

02. Follow the deciding steps

03. One club task with three kinds of work

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.